A Trained Iterative Classifier Reduces to Energy Gradient Descent: Recovering Closed-Form Dynamics from a Learned NLI System

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Abstract

We show that a trained neural update in an iterative NLI classifier can be removed and replaced with a closed-form energy gradient—with no retraining and no loss in performance (82.21% vs 82.05% on SNLI dev, $n = 9{,}842$).

We train an iterative classification head in which a hidden state evolves through multiple update steps before label readout, implemented as a learned MLP (1.2M parameters) trained end-to-end on SNLI. We find that this MLP is unnecessary. Replacing it with the analytical gradient of an energy function over anchor cosine similarities, $V(\boldsymbol{h}) = -\text{logsumexp}(\beta \cdot \cos(\boldsymbol{h}, A_k))$, produces identical performance without retraining. The trained network had converged to performing gradient descent on a simple, interpretable energy landscape—a result verified by ablation, not assumed by design.

We formalize this as three empirical patterns: (1) the initial state encodes the relational difference $h_0 = v_q - v_p$ (a design choice), (2) the representation space behaves as a logsumexp energy over anchor similarities (discovered), and (3) inference corresponds to gradient descent on that energy (discovered). Joint retraining under the recovered dynamics improves neutral recall by +6.1pp across five epochs and reduces head–dynamics disagreement from 55% to 4%. Trajectory analysis reveals a universal fixed-point collapse after ∼3 steps, identifying the primary accuracy bottleneck. Code and pretrained weights are publicly available.

1 Introduction

Standard neural classifiers map a representation h to logits in a single step. In this work, we consider an alternative: evolving h through a sequence of learned updates before classification, guided by anchor vectors that define label-specific basins in representation space.

The key finding is that the learned update is not required.

We verify this by deleting the trained MLP entirely and replacing it with the analytical gradient of an energy function—no retraining, no accuracy loss.

We train such a system on SNLI with a 1.2M-parameter MLP as the update rule. After training, we replace the MLP with the analytical gradient of an energy function defined over cosine similarities to the label anchors:

$$V(h) = -\log \sum_k \exp\big(\beta \cdot \cos(h, A_k)\big), \qquad h_{t+1} = h_t - \alpha \nabla_h V(h_t)$$

This parameter-free substitution produces the same accuracy on the full SNLI dev set (82.21% vs 82.05%, $n = 9{,}842$) without any retraining. The trained system had converged to performing gradient descent on a simple, interpretable energy landscape.

This was not designed. It was discovered by systematic ablation: removing the learned component and checking whether a closed-form alternative could reproduce the behavior. It could.

Context. The original Livnium system [Patil, 2026] introduced attractor-based classification with a bag-of-words encoder (76% on SNLI), using hand-designed cosine-radial forces with a 135° mismatch from the true cosine gradient—a non-conservative force field with no scalar potential. Upgrading to a jointly-trained BERT bi-encoder (82%) and analyzing the resulting dynamics revealed that the trained MLP had learned to correct this geometric inconsistency, producing behavior equivalent to gradient descent on $V(h)$.

Three empirical patterns. We formalize the discovered behavior as:

1. **Relational initialization** (design choice): $h_0 = v_q - v_p$
2. **Energy landscape** (discovered): $V(h) = -\text{logsumexp}\big(\beta \cdot \cos(h, A_E),\ \beta \cdot \cos(h, A_C),\ \beta \cdot \cos(h, A_N)\big)$
3. **Gradient dynamics** (discovered): $h_{t+1} = h_t - \alpha \nabla V(h_t)$

Patterns 2 and 3 were recovered empirically from a trained system, not assumed by design. Pattern 1 is a design choice that works well but is not fundamental—any relational encoding of the inputs should suffice.

Contributions.

1. Demonstration that a trained iterative classifier can be reduced to a parameter-free energy gradient with no accuracy loss (Section 5).
2. Characterization of the recovered dynamics as three empirical patterns (Section 4).
3. Joint retraining framework that aligns the classification head with the discovered dynamics (Section 6).
4. Tunnel test diagnostic that classifies error modes by trajectory analysis (Section 7.1).
5. Identification of a universal fixed-point collapse as the primary accuracy bottleneck (Section 7.2).

2 Background and Related Work

Energy-based models. The idea of classification via energy minimization has a long history. Hopfield networks [Hopfield, 1982] used energy-based attractor dynamics for content-addressable memory; modern Hopfield networks [Ramsauer et al., 2021] extend this with exponential storage capacity and connections to attention mechanisms. LeCun et al.'s energy-based learning framework [LeCun et al., 2006] provides a general formulation. Our work differs in that the energy function was not designed but *discovered* from trained dynamics.

Equilibrium and implicit models. Deep equilibrium models [Bai et al., 2019] compute representations as fixed points of implicit layers, and neural ODEs [Chen et al., 2018] parameterize continuous-time dynamics. Livnium uses explicit, finite-step dynamics with an interpretable geometric structure—the energy landscape can be written in closed form.

Prototype-based classification. Prototypical networks [Snell et al., 2017] classify by distance to class centroids. Livnium's anchors serve a similar role, but the classification involves iterative state refinement, not a single distance comparison. The anchors define an energy landscape, not just a nearest-neighbor rule.

Natural language inference. SNLI [Bowman et al., 2015] is the standard benchmark, with baselines from bag-of-words (∼80%) through decomposable attention [Parikh et al., 2016] (∼86%) to fine-tuned BERT [Devlin et al., 2019] (∼90%). Our work focuses on the classification head rather than the encoder; we demonstrate that the head itself contains discoverable physical laws.

3 The Livnium System

3.1 Architecture Overview

Given a premise p and hypothesis q, a BERT bi-encoder produces sentence vectors $\boldsymbol{v}_p, \boldsymbol{v}_q \in \mathbb{R}^{768}$. The initial state for the collapse dynamics is their difference (Law 1):

$$h_0 = \boldsymbol{v}_q - \boldsymbol{v}_p$$

Three learned anchor vectors $A_E, A_C, A_N \in \mathbb{R}^{768}$ (entailment, contradiction, neutral) define the geometry of the label space. Each anchor is unit-normalized during the forward pass.

3.2 Trained Collapse Rule (v1)

The original system evolves the state through $L = 6$ steps using:

$$h_{t+1} = h_t + \delta_\theta(h_t) - s_y \cdot D(h_t, A_y) \cdot \hat{\boldsymbol{n}}(h_t, A_y) - \beta \cdot B(h_t) \cdot \hat{\boldsymbol{n}}(h_t, A_N) \tag{1}$$

where δ_θ is a learned 2-layer MLP, $D(h, A) = 0.38 - \cos(h, A)$ is the divergence from the equilibrium ring, $\hat{\boldsymbol{n}}(h, A) = (h - A)/\|h - A\|$ is the Euclidean radial direction, and $B(h) = 1 - |\cos(h, A_E) - \cos(h, A_C)|$ is the boundary proximity. The equilibrium target for each anchor is a cosine-similarity ring at $\cos(h, A_y) = 0.38$, not the anchor point itself.

3.3 Classification Head

After L collapse steps, the refined state h_L is passed to a readout MLP (SNLIHead) along with auxiliary features: the original premise and hypothesis vectors $\boldsymbol{v}_p, \boldsymbol{v}_q$, cosine alignments, and geometric signals. The head outputs logits for three SNLI classes.

4 The Three Laws

We propose that the trained system's behavior is fully specified by three laws. Law 1 is a design choice; Laws 2 and 3 are empirical discoveries.

Definition 1 (Law 1: Relational State Formation)**.**

$$h_0 = \boldsymbol{v}_q - \boldsymbol{v}_p \tag{2}$$

The initial state encodes the direction of semantic change *from premise to hypothesis. Entailment pulls h_0 toward A_E; contradiction toward A_C; neutral is orthogonal.*

Definition 2 (Law 2: Energy Landscape)**.**

$$V(h) = -\log \sum_{k \in \{E,C,N\}} \exp\big(\beta \cdot \cos(h, A_k)\big) \tag{3}$$

The energy at any point h is the negative log-partition function of Boltzmann-weighted cosine similarities to the three class anchors. $V(h)$ assigns lower energy to points near an anchor, defining three semantic basins. The parameter β controls basin sharpness.

Definition 3 (Law 3: Collapse Dynamics)**.**

$$h_{t+1} = h_t - \alpha \nabla_h V(h_t) \tag{4}$$

The state evolves by gradient descent on the energy landscape over L steps. The gradient is computed analytically:

$$\nabla_h V(h) = -\sum_k w_k(\boldsymbol{h}) \cdot \nabla_h \cos(h, A_k) \tag{5}$$

where $w_k(h) = softmax\big(\beta \cdot \cos(h, A_k)\big)$ are the Boltzmann weights. The cosine gradient for each anchor is:

$$\nabla_h \cos(h, A) = \frac{A - h \cdot \cos(h, A)}{\|h\|} \tag{6}$$

Key property. The Boltzmann weights w_k ensure smooth interpolation near basin boundaries. At $\beta = 1$, the weights form a balanced softmax over anchors—preserving multi-basin gradient signal at boundaries. At $\beta \to \infty$, the dynamics reduce to argmax (nearest-anchor), losing boundary sensitivity.

5 Discovering the Equation of Motion

The three laws were not designed. They were discovered by a systematic comparison of the trained system with analytical alternatives.

5.1 Experimental Setup

We train a BERT bi-encoder jointly with the collapse engine on SNLI (549k training pairs, 9,842 dev pairs), achieving 82.06% dev accuracy. We then evaluate three collapse modes on the full dev set:

1. **Full**: The trained system (Eq. 1)—MLP δ_θ + anchor forces.
2. **No-delta**: Remove the learned MLP; keep only anchor forces.
3. **Grad-V**: Replace the entire collapse rule with Eq. 4—pure gradient descent on $V(h)$.

5.2 Result: The MLP Is Redundant

The key result is that removing the learned MLP does not degrade performance. Table 1 shows the analytical gradient of Eq. 3 matches the trained system. On the full dev set (n = 9,842), Grad-V actually *outperforms* the trained system by +0.16pp overall and +1.02pp on neutral recall. The learned MLP δ_θ, which constitutes 1.2M parameters and requires a forward pass at each collapse step, contributes at most 0.24pp. It is functionally dead.

Mode	Accuracy	E-recall	N-recall	C-recall
Original checkpoint ($\beta = 1$, full SNLI dev, $n = 9{,}842$)				
Full (δ_θ + forces)	82.05%	92.79%	71.16%	81.88%
No-delta (forces only)	81.81%	93.48%	68.62%	82.98%
Grad-V ($\beta = 1$, $\alpha = 0.2$)	**82.21%**	92.97%	**72.18%**	81.18%
Joint-retrained checkpoint ($\beta = 20$, $n = 2{,}000$)				
Full (δ_θ + forces)	82.35%	90.35%	71.64%	85.30%
No-delta (forces only)	82.20%	93.36%	69.87%	83.64%
Grad-V ($\beta = 20$, $\alpha = 0.2$)	**82.10%**	**94.42%**	**72.53%**	79.55%

Table 1: Gradient collapse comparison. Grad-V matches or exceeds the trained system on overall accuracy and neutral recall. The learned MLP δ_θ contributes $\leq 0.25\%$—it is functionally redundant. The system *is* a gradient flow on $V(h)$.

Interpretation. The trained MLP was approximating ∇V but introducing small distortions. The clean analytical gradient recovers and slightly exceeds the learned behavior. The energy $V(h)$ is the Boltzmann log-partition function over anchor similarities—a natural measure of semantic basin proximity. The system learned to perform gradient descent on this energy without being told to.

5.3 Geometric Robustness Under MLP Divergence

To test whether the grad-V result depends on the MLP being well-trained, we evaluated a second independent checkpoint where the MLP had begun to diverge from the anchor geometry. During fine-tuning from the primary checkpoint with a different random seed, the MLP weights drifted while the anchor structure remained intact ($\cos(A_E, A_C) = -0.188$, $\cos(A_C, A_N) = +0.118$). At this checkpoint the full system—MLP plus anchor forces—achieved only 70.66% accuracy, 11.4pp below the baseline. The MLP was actively degrading classification.

Mode	Accuracy	E-recall	N-recall	C-recall
Diverged-MLP checkpoint ($n = 9{,}842$)				
Full (δ_θ + forces)	70.66%	60.35%	58.36%	93.26%
No-delta (forces only)	81.95%	86.30%	72.64%	86.73%
Grad-V ($\beta = 2$, $\alpha = 0.05$)	**82.25%**	88.16%	**75.67%**	82.73%

Table 2: Grad-V evaluated on a checkpoint where the MLP has diverged. The full system (MLP + forces) collapses to 70.66%, while the analytical gradient of $V(h)$—which ignores the MLP entirely—recovers 82.25%. The anchor geometry outlasts the neural weights.

The analytical gradient recovered 82.25%, a gap of +11.59pp over the full system (Table 2). The anchor geometry, learned during the original training run, persisted unchanged; only the MLP had drifted. Because ∇V derives solely from the geometry and ignores the neural weights entirely, it cannot be degraded by MLP divergence. The MLP is not just redundant—it is a liability. The geometry is the classifier.

5.4 Stability Across β

We swept $\beta \in \{0.5, 1, 2, 5, 10, 20, 50\}$ at $\alpha = 0.1$. All values within $\beta \in [1, 50]$ produce accuracy within $\pm 1\%$ of the trained system, confirming that the energy landscape formulation is robust to the sharpness parameter. The result is not an artifact of hyperparameter tuning.

6 Joint Retraining Under Discovered Dynamics

The Grad-V discovery raises a natural question: if the dynamics are gradient descent on $V(h)$, can we retrain the system *explicitly* under these dynamics so that the classification head learns to trust the energy-based collapse?

6.1 Motivation: The TYPE-? Problem

At $\beta = 20$ (sharp basins), the gradient dynamics become highly decisive—states fall cleanly into the nearest anchor basin. However, the classification head (SNLIHead) was trained under the original $\beta = 1$ dynamics, where collapse produces a universal fixed point (Section 7.2). When we swap in $\beta = 20$ grad-V dynamics at test time, the head's predictions disagree with the dominant basin on 55.1% of errors—the head ignores the dynamics it was not trained on.

6.2 Method

We introduce **LiveniumJoint**, a joint training framework:

- BERT bi-encoder, standalone anchors $[A_E, A_C, A_N] \in \mathbb{R}^{3\times 768}$, and SNLIHead are trained together.
- Collapse uses differentiable grad-V dynamics (Eq. 4) with analytical gradient, staying in the computation graph.
- Loss combines cross-entropy with an anchor alignment term:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{\text{align}} \cdot \big(-\cos(h_L, A_y) \big) \tag{7}$$

The alignment loss directly penalizes disagreement between the final collapsed state and the correct anchor, forcing the head to be consistent with the dynamics.

6.3 Results

We train on 30,000 stratified samples (10k per class) for 5 epochs with $\beta = 20$, $\alpha = 0.05$, $\lambda_{\text{align}} = 0.3$, warm-starting from the BERT-joint checkpoint.

Joint retraining achieves its primary goal. Head–dynamics agreement rises from 29.7% to 73.9%, and neutral recall—the weakest class—improves monotonically across all five epochs, from 70.54% to 76.63% (+6.1pp). Overall accuracy is best at epoch 1 (82.79%), reflecting a trade-off: the alignment loss optimizes neutral recall at some cost to contradiction recall and overall accuracy. The Grad-V evaluation on the best checkpoint (Table 1, bottom) confirms that the analytical gradient still matches the full system, with N-recall of 72.53% exceeding the full model's 71.64%.

Metric	Before	Ep. 1*	Ep. 2	Ep. 3	Ep. 4	Ep. 5
Accuracy	81.95%	**82.79%**	82.48%	82.68%	82.43%	82.48%
Neutral recall	70.54%	74.50%	74.74%	75.98%	**76.63%**	76.51%
C recall	—	80.96%	79.62%	80.32%	78.83%	79.23%
Dyn/head agree	29.73%	72.52%	74.10%	73.29%	73.62%	73.88%
TYPE-? rate	7.84%	4.20%	3.98%	4.56%	4.50%	4.43%
$\cos(A_C, A_N)$	—	-0.031	-0.007	$+0.011$	$+0.024$	$+0.026$

Table 3: Full 5-epoch joint retraining (* = best checkpoint). Neutral recall improves *monotonically* across all epochs (70.5% → 76.6%, +6.1pp), peaking at epoch 4. Overall accuracy is best at epoch 1 (82.79%), reflecting a trade-off with C-recall. The final row shows $\cos(A_C, A_N)$ crossing zero at epoch 3 and reaching +0.026 by epoch 5—confirmed C–N anchor drift.

C–N anchor drift. $\cos(A_C, A_N)$ crosses zero at epoch 3 and reaches +0.026 by epoch 5. The C and N anchors become *positively* correlated under the alignment loss, and C-recall erodes from 80.96% to 78.83% correspondingly. This identifies a concrete regularization target: anchor orthogonality must be explicitly enforced during joint training. Adding a penalty $\lambda_{\text{orth}} \sum_{i \neq j} \cos(A_i, A_j)^2$ is the natural fix and a direct target for future work.

7 Trajectory Analysis

7.1 Tunnel Test: Classifying Error Modes

We develop a **tunnel test** diagnostic that traces the step-by-step trajectory of each collapse and classifies errors by *when* and *how* the dynamics fail. For each sample, we record the dominant anchor basin (highest cosine) at every step $t = 0, \ldots, L$.

Error types for neutral misclassifications:

- **TYPE-1** (bad h_0, Law 1 failure): h_0 starts in the wrong basin. The initial encoding does not place the state near the neutral anchor.
- **TYPE-2** (mid-diversion, Law 3 failure): h_0 starts correctly but the trajectory is diverted to a wrong basin during collapse.
- **TYPE-3** (boundary stall, Law 2 failure): the trajectory converges near a basin boundary or to the universal fixed point, never reaching a decisive basin.
- **TYPE-?** (head override): the dynamics deliver the correct dominant basin at step L, but the classification head overrides and predicts a different label.

7.2 The Universal Fixed Point

The dominant error mode at $\beta = 1$ is TYPE-3 (63.9%). Investigation reveals this is not boundary ambiguity but a more fundamental phenomenon: the collapse dynamics converge to a **universal fixed point** after approximately 3 steps, independent of the input.

At steps 4–6, nearly all samples—correct and incorrect—produce the same cosine alignment pattern:

$$\cos(h, A_E) \approx -0.129, \quad \cos(h, A_C) \approx -0.124, \quad \cos(h, A_N) \approx +0.172$$

	TYPE-1	TYPE-2	TYPE-3	TYPE-?
$\beta = 1$ (original)	14.4%	21.3%	**63.9%**	0.5%
$\beta = 20$ (grad-V)	2.8%	20.4%	21.7%	**55.1%**
$\beta = 20$ (joint)	—	—	—	**4.0%**

Table 4: Error type distribution across configurations. At $\beta = 1$, TYPE-3 (fixed-point convergence) dominates. At $\beta = 20$, dynamics become decisive but the head overrides them (TYPE-?). Joint retraining resolves the head–dynamics disconnect.

This fixed point is weakly neutral-biased ($\cos(h, A_N)$ is highest), explaining why neutral recall is better than chance (71%) but not as strong as entailment or contradiction. The collapse discards input-specific information after step 3—the final state is input-agnostic.

Energy signature. Correctly classified samples show mean energy descent $\Delta V = +0.046$ (ascending—moving toward lower-energy basins), while error samples show $\Delta V = -0.002$ (flat—stuck at the fixed point). The dynamics are contracting for correct predictions and stagnant for errors.

Implication. The accuracy bottleneck is not the classification head or the energy landscape—it is the fixed-point collapse. Input information is lost after three steps. Breaking this fixed point (e.g., via input-dependent residual forcing or adaptive step sizes) is the primary target for future work.

7.3 Basin Stability

Correct predictions occupy deeper attractor basins than incorrect ones. We measure stability by perturbing h_0 with Gaussian noise ($\sigma = 0.3$) across 20 trials per sample:

Group	Flip rate	Entropy
Correct ($n = 1{,}653$)	0.0017 ± 0.025	0.293
Incorrect ($n = 347$)	0.0068 ± 0.048	0.604
Ratio	**3.93**×	**2.06**×

Table 5: Basin stability: wrong predictions are 3.9× more likely to flip under perturbation and carry 2× higher entropy. Correctness correlates with basin depth—no calibration needed.

Per-class breakdown: entailment errors flip 7.18× more than correct entailment predictions, neutral errors 3.17×, contradiction errors only 1.82×. The low contradiction ratio indicates systematic geometric bias rather than boundary ambiguity—a structural feature of the anchor geometry.

8 Discussion

8.1 Why Does the MLP Learn ∇V?

The trained collapse rule (Eq. 1) uses cosine-magnitude forces with Euclidean-radial directions—a 135° mismatch from the true cosine gradient. Yet the *combined* effect of MLP + forces approximates gradient descent on $V(h)$. We hypothesize that the MLP learns a corrective residual that compensates for the geometric inconsistency of the hand-designed forces, effectively rotating the

force field toward the true gradient. The end-to-end training signal (cross-entropy on the final label) provides sufficient pressure to discover this correction.

8.2 What Is the Energy $V(h)$?

The energy $V(h) = -\text{logsumexp}(\beta \cdot \cos(h, A_k))$ is the negative log-partition function of a Boltzmann distribution over anchor cosine similarities. At any point h, it measures the aggregate proximity to all class anchors, weighted exponentially. The minima of V correspond to anchor positions; the saddle points define basin boundaries. At $\beta = 1$, the Boltzmann weights are a balanced softmax, preserving gradient signal at boundaries. At $\beta \to \infty$, $V(h) \to -\max_k \cos(h, A_k)$, recovering the Voronoi partition over anchors.

8.3 Accuracy Context

The system achieves 82–83% on SNLI, below the BERT-linear baseline (∼90%). The gap is attributable to two factors: (1) the bi-encoder architecture (premise and hypothesis encoded separately) loses cross-attention signal that a BERT cross-encoder exploits, and (2) the universal fixed-point collapse discards input information after 3 steps. The contribution of this work is not state-of-the-art accuracy but the *discovery of physical laws* governing a neural inference system—a result that holds regardless of absolute accuracy level.

8.4 Limitations

1. **Single dataset.** All results are on SNLI. Evaluation on MultiNLI, adversarial NLI, and non-NLI classification tasks would test generality.

2. **Single seed.** The gradient-flow result has been validated on the full dev set ($n = 9{,}842$) but from a single training run. Multi-seed validation is needed.

3. **Encoder dependence.** The laws are demonstrated with a BERT bi-encoder. Whether they hold for cross-encoders or non-BERT architectures is untested.

4. **Fixed-point bottleneck.** The universal fixed point limits accuracy. Breaking it requires architectural changes not yet validated.

5. **C–N anchor drift.** Joint retraining shows signs of C–N anchor convergence, which may limit contradiction recall.

9 Conclusion

We have shown that a trained attractor-based classification system for NLI is governed by three empirical laws: relational state formation, a logsumexp cosine energy landscape, and gradient-descent collapse dynamics. The central finding is that Laws 2 and 3 were *not designed*—they were recovered by analyzing what the trained MLP was doing. The learned residual δ_θ can be replaced by the analytical gradient ∇V with no accuracy loss on 9,842 samples.

Trajectory analysis via the tunnel test reveals the primary bottleneck: a universal fixed point that discards input information after three collapse steps. Joint retraining under the discovered dynamics reduces head–dynamics disagreement from 55% to 4% and improves neutral recall by +4pp.

The broader implication is that neural systems can develop interpretable physical laws through training—laws that are simpler, faster, and more transparent than the trained components they replace. Whether this phenomenon generalizes beyond attractor-based classification heads is an open question and a promising direction for future work.

Code and Models. https://github.com/chetanxpatil/livnium
https://huggingface.co/chetanxpatil/livnium-snli

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